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Center für Künstliche Intelligenz und Robotik
Technische Hochschule Würzburg-Schweinfurt

Fakultät Informatik und Wirtschaftsinformatik

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Portfolio

Computer Vision:

– Sheet 1 –

Portfolio 1 includes two exercises

Your goal is to integrate your knowledge of object detection methods, machine learning best practices, data quality, and deep metric learning concepts within a unified practical scenario. Each exercise is worth 15 points, and you must complete both exercises to achieve the full score.

Your submission should include:

- the trained detection model
- your implementation code (as a well-structured Python file following best coding practices)
- a short presentation of your solutions, covering
  - your training pipeline
  - evaluation results
  - an overview of the Assignment 2 solutions

This is a group assignment. Each team may consist of up to 5 members.

To enable an individual and differentiated assessment, your submission must contain a short contribution statement that clearly specifies which task(s) each group member was primarily responsible for. Multiple group members may collaborate on a task, but the individual contribution of each group member must still be clearly identifiable.

Each group member must be able to explain and defend their own contribution during the presentation and should have a general understanding of the overall solution.

Both solutions will be presented in a session, with 15 minutes allocated for the entire presentation.

Assignment 1:Object Detection for Automated Waste Sorting (15 Points)

Modern waste management systems increasingly rely on automated processes for identifying and separating different types of waste. Computer vision can support such systems by detecting waste objects and assigning them to appropriate material categories.

In this assignment, you will develop and evaluate an object detection model for identifying different categories of waste. You will use a Garbage Detection Dataset.

Your Tasks

You will select one object detection method covered in the lecture (e.g., YOLO, Faster R-CNN, SSD, etc.) and apply it to this dataset.

Task 1: Dataset Analysis and Designing the Training Pipeline (6 Points)

- Before training your model, perform an initial analysis of the provided dataset. Discuss relevant characteristics of the data with respect to the intended object detection task and explain how your observations affect the design of your training and evaluation pipeline
- Choose one object detection approach discussed in class and design a complete training pipeline following best practices, e.g.:
    - Split the dataset into training, validation, and test subsets
    - Apply appropriate data preprocessing and data augmentation techniques.
    - Select appropriate training parameters and strategies
    - Document and justify your pipeline clearly

Task 2: Model Training and Evaluation (5 Points)

- Train your detector using your defined pipeline on the provided dataset
- Evaluate your model on the test set using suitable object detection metrics
- Discuss your findings and interpret the metrics:
    - How well does your model detect the different waste categories?
    - For which types of objects or categories does the model perform well or poorly?
    - What are the strengths and limitations of your trained detector?

Task 3: Visualization and Error Analysis (4 Points)

- Visualize several example detections (2-3 per case):
    - Correct detections
    - Missed detections
    - Incorrect detections
- Display the corresponding bounding boxes and predictions on the test images
- Analyze the observed errors and discuss possible reasons for them. Based on your observations and evaluation results, discuss the limitations of your final model and possible approaches for improving its performance

Assignment 2: Deep Metric Learning for Waterfowl Detection and Conservation (15 Points)

Waste objects belonging to the same category can have considerably different visual appearances, while objects from different categories may appear visually similar. Deep Metric Learning provides a way to train models that represent such relationships in an embedding space.

In this assignment, you will apply Deep Metric Learning concepts to the waste detection scenario, **but only in a theoretical manner, i.e., without implementing or training the model.**

Your Tasks

Task 1: Loss Function and Sampling Strategy (8 Points)

Choose a suitable Deep Metric Learning loss function for learning an embedding representation of the waste categories.

Describe:

- the principle and objective of your selected loss function
- how training samples must be selected and combined for this loss
- how you would design an appropriate sampling strategy for the given dataset

Discuss how the sampling strategy can influence both the training process and the resulting embedding space.

Your proposed strategy should take the characteristics of the given dataset into account. Justify your decisions.

Task 2: Analysis and Use of the Learned Embedding Space (7 Points)

Assume that the Deep Metric Learning model from Task 1 has been successfully trained. Develop an approach for analyzing and using the resulting embedding space.

Your solution should address the following questions:

- How can the embedding representation be used to retrieve similar waste objects for a given query image?
- How could an unknown sample be assigned to one of the known waste categories using the learned embedding space?
- How can the structure of the learned embedding space be visualized and analyzed?
- How can the quality of the learned embedding space be evaluated quantitatively?

For each aspect, describe a suitable method and explain how its results should be interpreted.